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Course: Machine Learning
Project Report

Student Marks Prediction
Using Machine Learning

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Project Title	Student Marks Prediction Using Machine Learning	
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Roll No.	Name	Contribution
A216	Mehi Manswa Bhagat	Data loading, preprocessing, outlier handling, model training (all 5 models), evaluation, user input prediction, Streamlit dashboard
C264	Dipanshu Yadav	Research Work
C227	Aadi Bondre	Presentation and Word File Content

GitHub Link of Project:
https://github.com/student/student-marks-prediction

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1. Introduction

Student Marks Prediction

The Goal Schools collect a lot of data on students, but usually only find out someone is struggling *after* they fail a final exam. Our project uses machine learning to predict final scores early in the semester so teachers can step in and help.

What I Did

- **Predicted Grades:** Used data—like attendance, study hours, and sleep—to forecast final exam scores (0–100).
- **Tested Models:** Compared five machine learning models (Linear Regression, Decision Tree, Random Forest, KNN, and Lasso) to find the most accurate one.
- **Built an App:** Created an easy-to-use web dashboard using Streamlit so teachers can easily make predictions.

The Details I used the StudentPerformanceFactors dataset, which tracks 20 different student habits. All the coding, testing, and modeling was done in Python using Google Colab.

2. Literature Survey

Past Research

Predicting student grades isn't a new idea. Over the years, researchers have figured out a few key things:

- **Attendance is everything:** Early studies in the 90s proved an obvious point: showing up to class is the biggest factor in getting a good grade.
- **Better tech means better guesses:** When researchers started using machine learning, they found that advanced models (like Random Forest) make far fewer mistakes than basic ones.
- **It actually works in real life:** Purdue University once used a system that flagged struggling students with a warning light. Because teachers knew exactly who to help, the dropout rate fell by 21%.
- **Study time matters:** A big review in 2019 confirmed that attendance and study hours are the top two things that affect a score. Adding personal details, like a student's motivation, makes the predictions even smarter.

How My Project is Different Even though this topic has been studied a lot, my project adds a few new things:

- **Predicting exact scores:** Most old projects just try to guess if a student will "pass" or "fail." My model tries to predict their actual score out of 100.
- **Comparing models:** I am testing five different machine learning models side-by-side on the exact same data to see which one works best.
- **Making it usable:** A lot of research just stays in the code. I am building a real web app so teachers can type in a student's habits and instantly see their predicted score.

3. Methodology

How I Did It

The Data I loaded the StudentPerformanceFactors dataset using Python (Pandas). First, I checked to make sure there were no missing blank spaces or duplicate rows to mess things up. To make the data easier for the computer to read, I changed all the text-based columns into "category" formats.

Fixing Extreme Values (Outliers) Sometimes data has crazy, unrealistic numbers that can confuse a machine learning model. I used a math trick (the IQR method) to find these extreme numbers. Instead of deleting those students entirely, I used a method called "winsorization," which basically just caps those extreme numbers at a safe, normal limit.

My Testing Setup Even though I prepared a version of the dataset with all the features, I focused my main model training on the two most important factors: **Study Hours** and **Attendance**. I split the students up: 80% were used to teach the models, and the remaining 20% were kept hidden so I could test the models later.

The Models & Predictions I trained five different models on the data: Linear Regression, Decision Tree, Random Forest, KNN, and Lasso. Because models can sometimes do weird math, I added a rule to "clip" all their final guesses so they couldn't predict a grade lower than 0 or higher than 100. I even built a small script where a user can type in their own study hours (up to a maximum of 12) and attendance percentage to see what all five models predict they will score.

Checking the Math & Charts To figure out which model made the fewest mistakes, I calculated the Mean Absolute Error (MAE) and Mean Squared Error (MSE). I then created bar charts to compare these error scores side-by-side, and scatter plots to visually compare the models' guesses against the students' actual grades.

Building the Dashboard Finally, I built a live web app using Streamlit and hosted it using ngrok. The dashboard groups students into "Low," "Medium," and "High" performance categories. It includes a raw data view, scatter plots showing how study hours and attendance affect scores, a correlation heatmap, and sliders so users can filter students by their grades and attendance.

4. Result and Discussion

4.1 Model Performance Metrics

All five models were trained on the initial two-feature set (Hours_Studied, Attendance) and evaluated on the held-out test set. Results are presented in Table 4.1.

Table 4.1 – Model Evaluation Results

Model	MAE	MSE	Verdict
Random Forest	1.53	3.97	Moderate
Decision Tree	1.56	3.27	Good
Linear Regression ★ (Best)	**1.38**	**3.25**	**Best**
KNN Regressor	1.54	3.97	Moderate
Lasso Regression	1.40	3.31	Weak

The Results: Which Model Won?

With these new numbers, **Linear Regression** is the official winner! 🏆

To score the models, I looked at the Mean Absolute Error (MAE), which basically just tells us how many marks the model's guess was off by.

How They Did:

- **Linear Regression (The Best):** It had the lowest error score of all (an MAE of just 1.38). This means its predictions were incredibly accurate, usually missing the student's actual score by barely over 1 point.
- **Lasso Regression (Runner-Up):** Right on its heels! It scored a 1.40. Lasso uses almost the exact same math as Linear Regression, so it makes sense they tied for the top spot.
- **Random Forest & KNN (Middle of the Pack):** These models did okay (scoring around 1.53 and 1.54), but they weren't as sharp as the simpler math models.

- **Decision Tree (The Worst):** Just like before, this model came in last place with the highest error scores (MAE of 1.56). It still struggles to guess correctly for new students.

Why Did Linear Regression Win?

You might wonder why a "basic" model beat a fancy AI model like Random Forest.

It actually makes perfect sense! Since we are predicting grades based mostly on two things—**Study Hours** and **Attendance**—the relationship is very straightforward. The more you study and show up, the higher your grade goes. It's basically a straight line.

Linear Regression is designed specifically to draw simple, straight lines through data. The fancier models (like Random Forest) try to look for complex, hidden patterns that just don't exist here, which caused them to overthink the problem and make slightly worse guesses.

So, in this case, keeping the math simple was the smartest move!

5. Conclusion

The Big Takeaways (Conclusion)

This project proved that we can actually use everyday student habits to predict final grades with impressive accuracy before the test even happens.

- **Simple Math Won:** Because the relationship between studying, showing up, and getting good grades is pretty straightforward, **Linear Regression** ended up being the most accurate model. It usually guessed within 1.5 marks of a student's real score!
- **Overthinking Failed:** The **Decision Tree** model struggled the most. It tried to perfectly "memorize" the training students instead of learning the overall patterns, so it made the most mistakes on new students.
- **Ready for Teachers:** By building the Streamlit dashboard, this project isn't just a coding experiment—it’s a real, usable tool that teachers can click around on to identify and help struggling kids.

Where It Falls Short (Limitations)

- **I Kept It Basic:** For the main test, I focused heavily on just two factors (Study Hours and Attendance). Even though I prepared the data for all 20+ factors, I still need to fully test how the models perform when looking at everything at once.
- **The Data Was Too Perfect:** The Kaggle dataset was completely clean. Real-world school data is usually a mess, with missing information and typos that would require a lot more cleanup.
- **People Change:** The model just looks at averages. It doesn't account for a student who might slack off in September but suddenly starts studying incredibly hard right before the final.

What I Want to Do Next (Future Enhancements)

- **Test Everything:** I want to run the models using every single piece of data available to see if the predictions get even sharper.
- **Smarter AI:** I'd like to try advanced deep learning models that can track a student's timeline and see if their habits are improving or declining over the semester.

- **Wearable Tech:** It would be fascinating to add data from smartwatches—like daily stress levels or actual sleep tracking—to see how that impacts test scores.
- **A Real School System:** I want to upgrade the web app into a full, school-wide software platform where different teachers and principals have their own secure logins.
- **Explain the "Why":** I want to add a feature that doesn't just give a predicted score, but actually tells the teacher *exactly why* the model gave that score (for example: "This score is low specifically because attendance dropped below 70%").

6. References

Where I Got My Information (Sources)

The Data

- **Kaggle:** I downloaded the main StudentPerformanceFactors dataset directly from Kaggle.

The Tech & Tools I Used

- **Python & Pandas (2010):** I used Pandas to organize and clean up all the data.
- **Scikit-learn (2011):** This is the main Python library I used to actually build and test the machine learning models.
- **Random Forest (2001):** I referenced the original research paper that invented the Random Forest model to understand how it works.
- **Streamlit (2023):** I used Streamlit's official guides to help me code and build the final web dashboard.

Past Research I Looked At To make sure I was building my project correctly, I read up on what other researchers have done in the past:

- **The Basics of Predicting Grades:** I looked at older papers from 2004 and 2010 that first started using machine learning to predict how well students would do in class.

2004 Paper: Predicting students' performance in distance learning using machine learning techniques (Kotsiantis et al.)Link to Study (DOI: 10.1080/08839510490442058)

2010 Paper: *Educational data mining: A review of the state-of-the-art* (Romero & Ventura) Link to Study (DOI: 10.1109/TSMCC.2010.2053532)

- **Real-World Success:** I read a 2012 study about Purdue University's "Course Signals" program to see how predicting grades actually helps lower dropout rates in real life.

2012 Purdue Study: *Course Signals at Purdue: Using learning analytics to increase student success* (Arnold & Pistilli)Link to Study (DOI: 10.1145/2330601.2330666)

- **Finding the Best Model:** I referenced studies from 2012 and 2019 that proved why advanced models (like Random Forest) are so good at predicting student success, and why things like study hours matter so much.

2012 Random Forest Study: *Mining educational data to reduce dropout rates of engineering students* (Pal) Link to Study (DOI: 10.5815/ijieeb.2012.02.01)

2019 Study on Habits/Study Hours: *Educational data mining: Predictive analysis of academic performance of public school students* (Fernandes et al.) Link to Study (DOI: 10.1016/j.jbusres.2018.02.012)